

VALIDATION OF LAMP ASSAY FOR *Klebsiella aerogenes* DETECTION IN THREE VEGETABLE SPECIES

Marija Pavlović¹, Mila Djisalov², Ljiljana Šašić Zorić², Ljiljana Janjušević², Željko D. Popović¹, Ivana Gadjanski²

¹Department of Biology and Ecology, Faculty of Sciences, University of Novi Sad, Trg Dositeja Obradovića 3, 21000 Novi Sad, Serbia;

²BioSense Institute, University of Novi Sad, Dr Zorana Đinđića 1, 21000 Novi Sad, Serbia

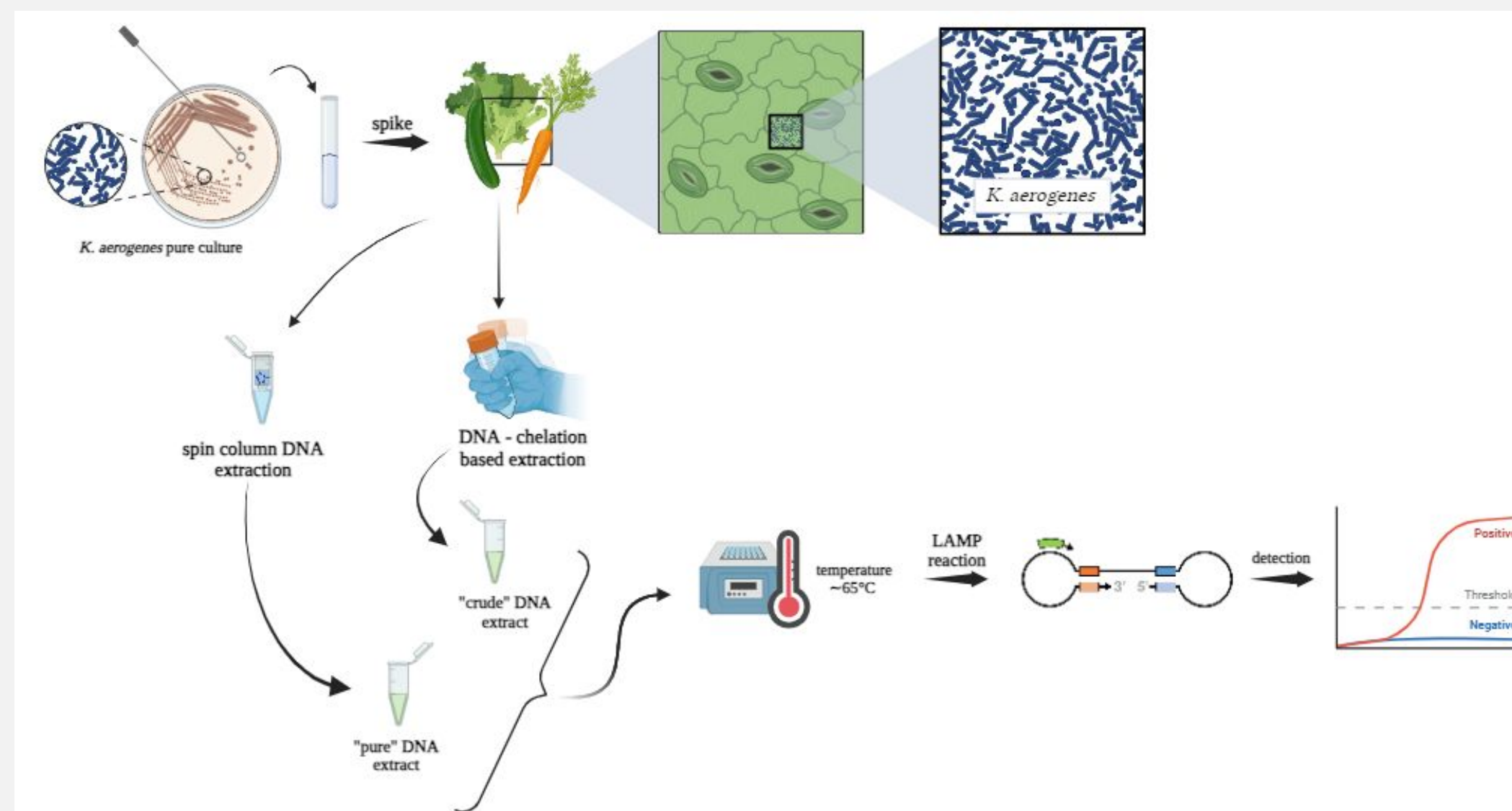


Figure 1. Graphical abstract of the experiment. Artificial contamination (spiking) of vegetables followed by DNA extraction and LAMP analysis.

MATERIALS & METHODS

We selected three distinct vegetable types (carrot, cucumber, and lettuce) that were previously tested for natural contamination by *K. aerogenes*. Since the test results came in negative, the tested vegetables were then artificially contaminated (spiked) with a bacterial suspension containing 1.5×10^8 CFU/mL of *K. aerogenes* in the final sample volume. DNA extraction from the samples was performed using two approaches: Chelex approach for fast extraction to obtain “crude” DNA extract, and Plant/Fungi DNA Isolation Kit (Norgen Biotek Corp., Canada) for spin column extraction to obtain “pure” DNA extract (**Fig 1**). The total DNA extracted from both methods was subsequently diluted to a final concentration of 1 ng/μL.

Final samples were then used for LAMP detection using WarmStart LAMP Kit (New England Biolabs) for Real-Time LAMP. Real-Time LAMP reactions were done on the Genie® III (OptiGene), an instrument for isothermal nucleic acid amplification. LAMP assay was conducted at 65°C for 30 minutes. In this study, we used LAMP primers that targeted the *K. aerogenes* *HDC* gene, and which were previously developed our laboratory.

INTRODUCTION

Klebsiella aerogenes is a foodborne pathogen that contaminates fruits and vegetables. Detecting it in a timely manner is essential to prevent its transmission through contaminated food sources. In this regard, the LAMP method proved to be a robust, highly sensitive and highly specific method for real-time pathogen detection, and thus method of choice for detection of *K. aerogenes*. In order to check efficiency of LAMP methodology with fast DNA extraction we conducted LAMP-based *K. aerogenes* detection in vegetables spiked with bacteria.

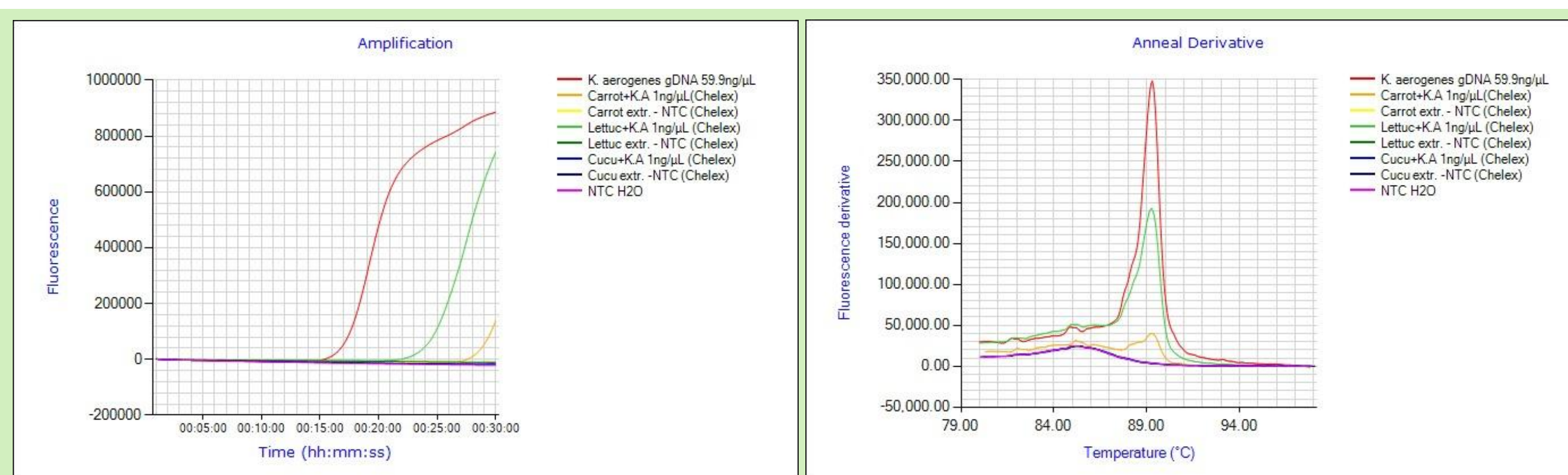


Figure 2a. Amplification signals of Real-Time LAMP reaction for the Chelex extracts.

Figure 2b. Melting curves of Real-Time LAMP reaction for the Chelex extracts.

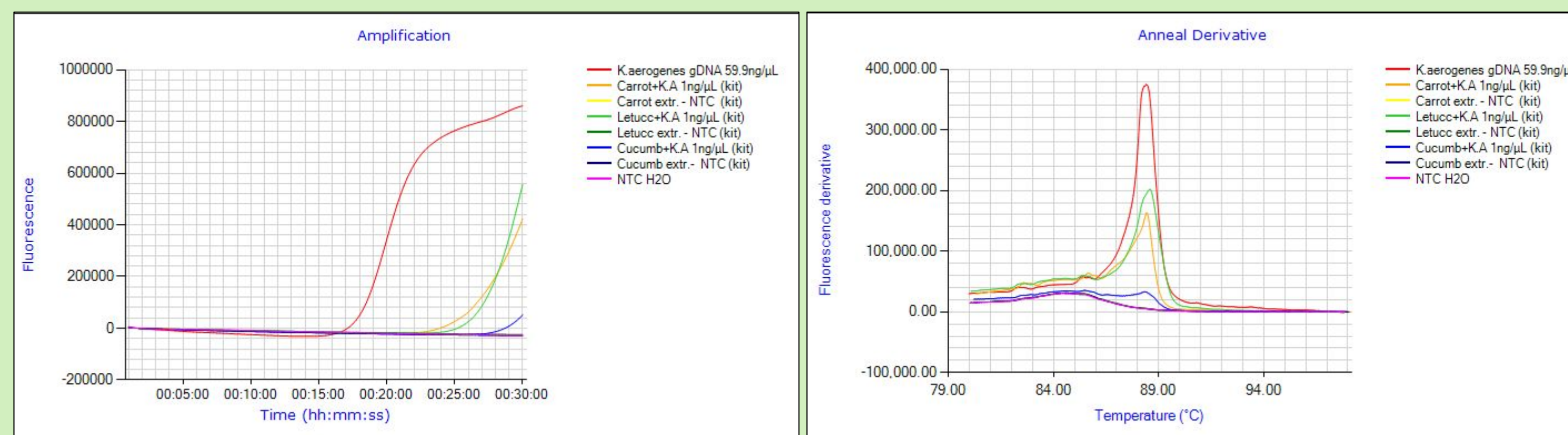


Figure 3a. Amplification signals of Real-Time LAMP reaction for the spin column extracts.

Figure 3b. Melting curves of Real-Time LAMP reaction for the spin column extracts.

RESULTS

The results indicated successful detection of *K. aerogenes* by LAMP using total DNA from both extraction approaches. DNA isolated with a commercial kit showed better results compared to the DNA obtained by the Chelex method. For the commercial kit, all three spiked samples were detected: spiked carrot and spiked lettuce around 10 minutes later compared to the positive control, and around 15 minutes later for spiked cucumber (**Fig 3a**).

On the other hand, for the samples obtained with the Chelex method, two of three samples were detected (spiked lettuce and spiked carrot), while the spiked cucumber sample was not detected (**Fig 2a**). This can be related to the anatomy of the sampled cucumber tissue, which contained the exocarp and mesocarp. *K. aerogenes* contamination usually occurs on the surface of the cucumber (exocarp) and doesn't go further into the tissue (mesocarp). The exocarp contains resin and is therefore harder to homogenize.

Regarding the non spiked extracts, the absence of the amplification signal implied that contamination by pathogenic *K. aerogenes* was not present before the spiking. As for the difference observed in the melting curves between the samples, the low peaks of the curve can be a consequence of the low amount of product obtained in the shorter amplification time for these samples (**Fig 2b and 3b**). The overall lower efficiency of the Chelex approach can be related to the fact that fast DNA extracts often contain nucleic acid amplification inhibitors.

CONCLUSION

- ✓ The presented research highlights the remarkable potential of the LAMP methodology that can be efficiently combined with fast DNA extraction methods for in-field detection of pathogens.
- ✓ Improvements for the Chelex method could be made in light of sample selection, different sampling approaches, homogenization methods (specifically the intensity and duration of homogenization) and removal of potential reaction inhibitors from the obtained “crude” extracts.
- ✓ The commercial kit showed better compatibility with LAMP for sample preparation and the Chelex method has shown great potential for point-of-need use.